at blinding speed. You need to picture the whole of the universe squeezing together into an ever shrinking cosmic mass. As in Chapter 3, from a maximum size of many billions of light-years, the universe will shrink to millions of light-years, every moment gaining speed as everything is crushed together to the size of a single galaxy, and then to the size of a single star, a planet, and down to the size of an orange, a pea, a grain of sand, and further, according to general relativity, to the size of a molecule, an atom, and in a final inexorable cosmic crunch to no size at all. According to conventional theory, the universe began with a bang from an initial state of zero size, and if it has enough mass, it will end with a crunch to a similar state of ultimate cosmic compression.

But when the distance scales involved are around the Planck length or less, quantum mechanics invalidates the equations of general relativity, as we are now well aware. We must instead make use of string theory. And so, whereas Einstein's general relativity allows the geometrical form of the universe to get arbitrarily small—in exactly the same way that the mathematics of Riemannian geometry allows an abstract shape to take on as small a size as the intellect can imagine—we are led to ask how string theory modifies the picture. As we shall now see, there is evidence that string theory once again sets a lower limit to physically accessible distance scales and, in a remarkably novel way, proclaims that the universe cannot be squeezed to a size shorter than the Planck length in any of its spatial dimensions.

Based on the familiarity you now have with string theory, you might be tempted to hazard a guess as to how this comes about. After all, you might argue that no matter how many points you pile up on top of each other—point particles that is—their combined volume is still zero. By contrast, if these particles are really strings, collapsed together in completely random orientations, they will fill out a nonzero-sized blob, roughly like a Planck-sized ball of entangled rubber bands. If you made this argument, you would be on the right track, but you would be missing significant, subtle features that string theory elegantly employs to suggest a minimum size to the universe. These features serve to emphasize, in a concrete manner, the new stringy physics that comes into play and its resultant impact on the geometry of spacetime.